

# Kenneth Loi

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## EDUCATION

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| <b>Ph.D., Biophysics</b><br><i>University of California, Berkeley</i>                | 2026–present |
| <b>B.A., Molecular and Cell Biology</b><br><i>University of California, Berkeley</i> | 2024         |

## RESEARCH INTERESTS

RNA-guided systems; phage defense; biomolecular evolution; CRISPR biology; structural bioinformatics; AI for biology

## RESEARCH EXPERIENCE

|                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                               |                           |
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| <b>Graduate Researcher</b><br><i>Doudna Lab, UC Berkeley</i><br><i>Previously undergraduate researcher and research technician</i>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                            | 2026–present<br>2023–2026 |
| <ul style="list-style-type: none"><li>Co-first author of the VIPR discovery study, published in <i>Science</i> (2026); identified a phage-encoded RNA-guided DNA recognition system with a noncontiguous targeting code.</li><li>Used genomic language models and small RNA sequencing to discover a novel guide RNA for VIPR.</li><li>Developed structural bioinformatics and comparative genomics pipelines to identify divergent RNA-guided systems and infer their evolutionary relationships.</li><li>Designed and executed plasmid-based functional assays for programmable repression, phage-defense genetics, and guide-target specificity.</li></ul> |                           |
| <b>Undergraduate Researcher</b><br><i>Murthy Lab, UC Berkeley</i>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                             | 2021–2023                 |
| <ul style="list-style-type: none"><li>Tested lipid nanoparticle (LNP) formulations for nucleic acid delivery.</li><li>Investigated mRNA chemical modifications to enhance transcript stability and translation efficiency.</li><li>Collaborated on projects delivering gene editing effectors to target tissues.</li></ul>                                                                                                                                                                                                                                                                                                                                    |                           |
| <b>Summer Undergraduate Researcher (SR-EIP)</b><br><i>Jones Lab, Brown University</i>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                         | 2021                      |
| <ul style="list-style-type: none"><li>Helped develop the Human Neocortical Neurosolver (HNN): a software for biophysical modeling of MEG/EEG potentials of neural circuits</li><li>Leveraged HNN to study possible mechanisms driving Alzheimer’s-associated MEG/EEG readings</li></ul>                                                                                                                                                                                                                                                                                                                                                                       |                           |

## PUBLICATIONS

- Yoon PH, **Loi KJ**, Zhang ZT, Docter TA, Lopez SC, Langeberg CJ, Ur-Rehman MM, Vohra K, Zhou Z, Esain-Garcia I, et al. A noncontiguous code for RNA-guided DNA recognition at the origin of CRISPR-Cas. *Science* 393(6817):1230–1235, 2026. doi: 10.1126/science.aei0498. **Co-first author.**
- Yoon PH, Docter TA, Zhang ZT, **Loi K**, Lopez SC, Valentin-Alvarado LE, Tuck OT, Brohawn SG, Doudna JA. VIPR RNA-guided DNA recognition by noncontiguous geometric triplex formation. *Science* 393(6817):1236–1240, 2026. doi: 10.1126/science.aei3472.
- Skopintsev P, Esain-Garcia I, DeTurk EC, Yoon PH, Zhou Z, Weiss T, Kamalu M, Chamraj A, **Loi KJ**, Langeberg CJ, et al. Structure and evolution-guided design of minimal RNA-guided nucleases. *Science* 393(6808):313–318, 2026. doi: 10.1126/science.aed6123.
- Tuck OT, Hu JJ, Lopez SC, Adler BA, O’Brien CE, Hsieh K, Meredith C, **Loi KJ**, Yoon PH, Doherty EE, et al. Recurrent acquisition of nuclease-protease pairs in antiviral immunity. *Science* 391(6781):195–201, 2026. doi: 10.1126/science.aea8769.

- Doherty EE, Adler BA, Yoon PH, Hsieh K, **Loi K**, Armbruster EG, Lahiri A, Bolling CS, Wilcox XE, Akkati A, et al. A miniature CRISPR–Cas10 enzyme confers immunity by inhibitory signalling. *Nature* 647(8091):997–1004, 2025. doi: 10.1038/s41586-025-09569-9.
- Yoon PH, Zhang Z, **Loi KJ**, Adler BA, Lahiri A, Vohra K, Shi H, Rabelo DB, Trinidad M, Boger RS, et al. Structure-guided discovery of ancestral CRISPR-Cas13 ribonucleases. *Science* 385(6708):538–543, 2024. doi: 10.1126/science.adq0553.
- Yoon PH, Skopintsev P, Shi H, Chen L, Adler BA, Al-Shimary M, Craig RJ, **Loi KJ**, DeTurk EC, Li Z, et al. Eukaryotic RNA-guided endonucleases evolved from a unique clade of bacterial enzymes. *Nucleic Acids Research* 51(22):12414–12427, 2023. doi: 10.1093/nar/gkad1053.
- Jas M, Thorpe R, Tolley N, Bailey C, Brandt S, Caldwell B, Cheng H, Daniels D, Pujol CF, Khalil M, Kanekar S, Kohl C, Kolozsvári O, Lankinen K, **Loi K**, et al. HNN-core: A Python software for cellular and circuit-level interpretation of human MEG/EEG. *Journal of Open Source Software* 8(92):5848, 2023. doi: 10.21105/joss.05848.
- Sharma R, William BE, **Loi K**, Murthy N. Efficient delivery of nucleic acids by lipid nanoparticles to the liver, spleen, and brain. *Molecular Therapy* 31(4):286, 2023.

## PRESENTATIONS & POSTERS

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| <b>A noncontiguous code at the origin of CRISPR</b>                                           | 2026        |
| <i>CRISPR 2026 — Talk</i>                                                                     | <i>June</i> |
| <b>Scalable Structural Homology Pipeline Reveals Hidden Protein Relationships</b>             | 2025        |
| <i>Cold Spring Harbor Genome Engineering Meeting — Poster</i>                                 |             |
| <b>Structure-guided discovery of ancestral Cas13s</b>                                         | 2024        |
| <i>Sternberg Lab Seminar; Columbia University — Talk</i>                                      |             |
| <b>Biophysical modeling of Alzheimer’s disease mediated changes in MEG alpha oscillations</b> | 2021        |
| <i>Annual Biomedical Research Conference for Minority Students — Poster</i>                   |             |

## AWARDS & FELLOWSHIPS

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| <b>NSF Graduate Research Fellowship Program</b>                               | 2026 |
| <i>Honorable Mention</i>                                                      |      |
| <b>UC Berkeley Rose Hills Fellow</b>                                          | 2022 |
| <b>The Leadership Alliance Summer Research - Early Identification Program</b> | 2021 |

## TEACHING & MENTORSHIP

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| <b>SPLASH: Molecular Matrix: Weaving Biology and Machine Learning Together</b>        | 2024 |
| <i>Taught high school students about applications of machine learning in biology.</i> |      |
| <b>SPLASH: Gene editing for human health. A CRISPR view and beyond</b>                | 2022 |
| <i>Taught high school students how CRISPR is impacting human health</i>               |      |

## TECHNICAL SKILLS

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**Wet-lab:** Molecular cloning, protein purification, mammalian cell culture (transfection, maintenance), plate-reader assays.

**Bioinformatics:** HMMER, IQ-TREE, AlphaFold, Foldseek

**Programming:** Python, Bash, LaTeX.